

Reinforcement Learning-Based Intelligent Control of Lower Limb Exoskeletons for Personalized Gait Rehabilitation

Yiyang He¹, Ziwei Wang¹, and Tao Xue²

¹Lancaster University, Lancaster, United Kingdom

²Tsinghua University, Beijing, China

Abstract

Lower limb exoskeletons hold great promise for gait rehabilitation, yet conventional PID and impedance controllers suffer from response lag, inability to adapt to individual gait abnormalities, and unnatural assistance. Recent breakthroughs show that reinforcement learning (RL) policies trained in simulation can achieve up to 24.3% metabolic cost reduction on real hardware. This work proposes a dual-layer RL framework: the first layer builds patient-specific pathological gait models using the MyoSuite musculoskeletal simulation (80 muscles) with a metabolic energy reward; the second layer trains an anticipatory assistive policy minimizing muscular effort with near-zero-lag torque delivery. A curriculum-based progressive reward guides pathological gait toward healthy patterns, while instrumented insole sensors provide real-time gait phase detection. The framework aims to advance exoskeletons from passive tools toward intelligent rehabilitation agents.

1. Introduction and Motivation

Millions of patients worldwide require gait rehabilitation following stroke, spinal cord injury, or other neuromotor disorders. Robotic lower limb exoskeletons provide repeatable and quantifiable gait training [1], yet most systems rely on conventional PID or impedance controllers that track predefined trajectories [2]. These exhibit three limitations: (1) response lag, (2) inability to accommodate patient-specific abnormalities, and (3) fighting residual voluntary motion [3].

Deep RL has demonstrated remarkable progress in musculoskeletal control [5] and legged robotics [4]. Luo et al. [6] achieved a record 24.3% metabolic cost reduction with a hip exoskeleton controller trained entirely in simulation without human-in-the-loop experiments, then validated on real hardware. Complementarily, Slade et al. [7] demonstrated personalized outdoor exoskeleton assistance with 17%±5% energy reduction using data-driven human-in-the-loop optimization. Together, these confirm that data-driven and learning-based approaches now outperform hand-tuned classical methods for exoskeleton control.

This work presents an RL framework addressing four questions: (Q1) How to model patient-specific pathological gait? (Q2) How to achieve anticipatory near-zero-lag torques? (Q3) How to progressively guide gait toward healthy patterns? (Q4) How to perceive gait phase in real time?

2. Problem Statement and Methodology

The proposed framework consists of five modules (Fig. 1).

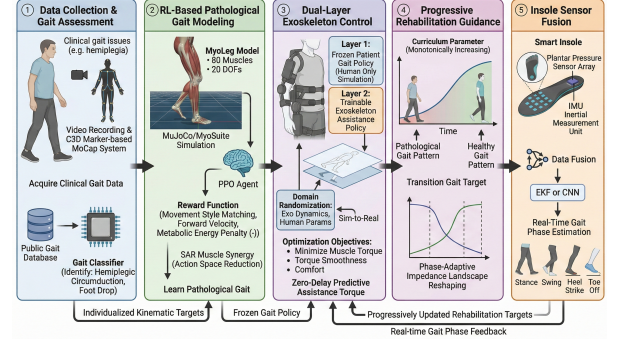


Figure 1: Overview of the proposed RL-based exoskeleton control and rehabilitation framework.

2.1 Data Acquisition and Gait Assessment

Clinical gait data are collected via video recording and C3D motion capture. A classifier trained on public databases discriminates disease-specific features (hemiplegic circumduction, foot drop), ensuring simulations faithfully reproduce patient-specific pathological movement.

2.2 RL-Based Pathological Gait Modeling

Patient-specific gait is modeled using *MyoSuite/MyoLeg* [5], a MuJoCo-based simulation running orders of magnitude faster than OpenSim. MyoLeg features 80 validated muscle actuators across 20 DOF. A PPO agent learns a walking policy matching pathological kinematics, building on the RL-driven neuromechanical simulation paradigm reviewed by Song et al. [8], with a metabolic energy reward based on the Umberger–Uchida model [9]:

$$R_{\text{gait}} = w_1 R_{\text{style}} + w_2 R_{\text{vel}} - w_3 C_{\text{metabolic}} \quad (1)$$

where R_{style} penalizes kinematic deviations, R_{vel} rewards forward progression, and $C_{\text{metabolic}}$ penalizes excessive muscle activation. Schumacher et al. [10] showed that biologically plausible reward functions (adaptive effort minimization) can produce natural walking across musculoskeletal models (up to 90 muscles) *without motion capture demonstrations*—suggesting that demonstration-free RL is viable for modeling patient-specific gait where reference data may be unavailable. To address exploration in the 80-D muscle space, we adopt SAR [11], leveraging biological muscle synergies to reduce action dimensionality.

2.3 Dual-Layer Exoskeleton Control

The *dual-layer RL architecture* decouples gait modeling from assistive control, inspired by Luo et al. [6], who coupled a full-body 208-muscle musculoskeletal model with a hip exoskeleton and trained via PPO in ~8 hours on one

GPU. Our first layer simulates the patient without assistance. The second layer trains an assistive policy atop the frozen gait policy:

$$R_{\text{exo}} = -\alpha \sum_i \tau_i^{\text{muscle}} + \beta R_{\text{smooth}} - \gamma \|\ddot{\tau}^{\text{exo}}\| \quad (2)$$

minimizing muscular effort while penalizing torque jerk ($\ddot{\tau}^{\text{exo}}$, the second time-derivative of exoskeleton torque) to ensure smooth, comfortable assistance profiles. By learning to *anticipate* intended motion, the architecture achieves near-zero-lag torque delivery. Domain randomization of exoskeleton dynamics and musculoskeletal parameters enables robust sim-to-real transfer [6].

2.4 Progressive Rehabilitation Guidance

Following the assist-as-needed (AAN) paradigm [12], the gait target progressively shifts from pathological toward healthy gait via *curriculum learning*:

$$G_{\text{target}}(t) = (1 - \lambda(t)) \cdot G_{\text{pathological}} + \lambda(t) \cdot G_{\text{healthy}} \quad (3)$$

where $\lambda(t) \in [0, 1]$ increases monotonically. The controller simultaneously reshapes phase-dependent impedance landscapes [13], providing more assistance where patients struggle and less where they demonstrate competence, avoiding abrupt corrections.

2.5 Insole Sensor Fusion

Accurate real-time gait phase estimation is essential for timely torque delivery. Medrano et al. [14] demonstrated that EKF-based kinematic phase estimation using onboard joint encoders and IMUs can achieve $4.8 \pm 2.4\%$ RMSE on treadmill and comparable accuracy on uneven outdoor terrain, while Tang et al. [15] reported sub-20 ms latency via IMU-driven multi-resolution neural networks. Building on these IMU-based approaches, our framework further incorporates instrumented insoles with plantar pressure sensor arrays (Fig. 2) to provide complementary ground contact information, enabling robust detection of stance/swing transitions and heel-strike/toe-off events across diverse walking surfaces.

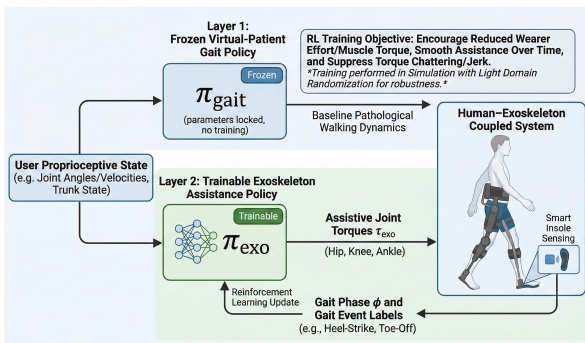


Figure 2: Dual-layer RL control architecture with real-time gait phase input from instrumented insoles.

3. Expected Outcomes

The framework targets four measurable outcomes: (1) Patient-specific gait simulations with muscle activations correlated with clinical EMG, following KINESIS [16]

(290-muscle physiological plausibility); (2) Significant metabolic cost reduction via sim-to-real transfer (reference: 24.3% [6] on healthy subjects; pathological targets to be established through simulation); (3) Gait symmetry improvement over progressive sessions, with online RL adaptation converging within minutes based on analogous prosthesis tuning results [17]; and (4) Gait phase RMSE of approximately 5%, consistent with current EKF and deep learning methods [14]. Validation proceeds from MuJoCo simulation through pilot hardware deployment.

4. Conclusion and Significance

This work contributes a unified RL-based pipeline spanning gait diagnosis, intelligent assistance, and progressive rehabilitation for lower limb exoskeletons. The contributions include: (i) personalized musculoskeletal gait modeling with metabolic rewards on MyoSuite, (ii) a dual-layer cooperative control architecture with sim-to-real transfer via domain randomization, and (iii) curriculum-based rehabilitation rewards aligned with clinical AAN principles. While surveys [3, 18] note deep RL remains under-validated clinically, our framework directly targets this translational gap through patient-specific pathological modeling. The vision is a complete chain from gait diagnosis → intelligent assistance → rehabilitation guidance, transforming exoskeletons into intelligent rehabilitation therapists.

References

- [1] B. Chen, H. Ma, L.-Y. Qin, et al., "Recent developments and challenges of lower extremity exoskeletons," *J. Orthop. Transl.*, 5:26–37, 2016.
- [2] R. Baud, A. R. Manzoori, A. Ijspeert, and M. Bouri, "Review of control strategies for lower-limb exoskeletons to assist gait," *J. NeuroEng. Rehabil.*, 18:119, 2021.
- [3] O. Coser, C. Tamantini, P. Soda, and L. Zollo, "AI-based methodologies for exoskeleton-assisted rehabilitation: a review," *Front. Robot. AI*, 11, 1341580, 2024.
- [4] J. Hwangbo, J. Lee, A. Dosovitskiy, et al., "Learning agile and dynamic motor skills for legged robots," *Sci. Robot.*, 4(26), eadu5872, 2019.
- [5] V. Caggiano, H. Wang, G. Durandau, A. Sartori, and V. Kumar, "MyoSuite – a contact-rich simulation suite for musculoskeletal motor control," in *Proc. L4DC*, PMLR 168, 2022.
- [6] S. Luo, M. Jiang, S. Zhang, et al., "Experiment-free exoskeleton assistance via learning in simulation," *Nature*, 630:353–359, 2024.
- [7] P. Slade, M. J. Kochenderfer, S. L. Delp, and S. H. Collins, "Personalizing exoskeleton assistance while walking in the real world," *Nature*, 610:277–282, 2022.
- [8] S. Song, L. Kidziński, X. B. Peng, et al., "Deep RL for modeling human locomotion control in neuromechanical simulation," *J. NeuroEng. Rehabil.*, 18:126, 2021.
- [9] T. K. Uchida, J. L. Hicks, C. L. Dembia, and S. L. Delp, "Stretching your energetic budget: how tendon compliance affects the metabolic cost of running," *PLoS ONE*, 11(3), e0150378, 2016.
- [10] P. Schumacher, T. Geijtenbeek, V. Caggiano, V. Kumar, S. Schmitt, G. Martius, and D. F. B. Haeufle, "Emergence of natural and robust bipedal walking by learning from biologically plausible objectives," *iScience*, 28(4), 112203, 2025.
- [11] D. Berg, V. Caggiano, and V. Kumar, "SAR: Generalization of physiological agility via synergistic action representation," in *Proc. RSS*, 2023.
- [12] W. Zhang, K. Li, K. Nolan, and D. Zanotto, "Adaptive assist-as-needed control based on actor-critic RL," in *Proc. IROS*, pp. 4066–4071, 2019.
- [13] W. Zhang and D. Zanotto, "Shaping individualized impedance landscapes for gait training via RL," arXiv:2109.02109, 2021.
- [14] R. L. Medrano, G. C. Thomas, C. G. Keais, E. J. Rouse, and R. D. Gregg, "Real-time gait phase and task estimation for controlling a powered ankle exoskeleton on extremely uneven terrain," *IEEE Trans. Robot.*, 39(3):2170–2182, 2023.
- [15] Y. Tang, M. Shushtari, and A. Arami, "IMU-based real-time gait phase estimation using multi-resolution neural networks," *Sensors*, 24(8), 2390, 2024.
- [16] M. K. Simos, A. S. Chiappa, and A. Mathis, "KINESIS: Motion imitation for human musculoskeletal locomotion," arXiv:2503.14637, 2025.
- [17] Y. Wen, J. Si, A. Brandt, X. Gao, and H. H. Huang, "Online RL control for personalization of a robotic knee prosthesis," *IEEE Trans. Cybern.*, 50(6):2346–2356, 2020.
- [18] M. Sharifi, S. Tripathi, H. Chen, K. Zhang, and M. Tavakoli, "RL methods for assistive and rehabilitation robotic systems: a survey," *IEEE Trans. SMC: Systems*, 55:4534–4551, 2025.